

CLAIM - Alpha as a 21|30 Closure Gate v38.4

Primary box formula first; ledger links explain where each factor enters.

The formula below is the primary claim, not a later numerical add-on. The rest of the note is the ledger explaining where each factor enters.

Primary claim: α_{RT} is the closure margin

$$\alpha_{RT} = \frac{\pi}{30 \cdot 1260} (3 \cdot 30 - \Delta)$$

Expanded correction

$$\Delta = 2 + \frac{2}{10} - \frac{1}{10 \cdot 39} - \frac{20/21}{10 \cdot 39 \cdot 30 \cdot 7}$$

Factor map

Factor	Meaning
pi	circular / half-turn geometry scale
30	C30 carrier/readout grid
1260	first full active closure
3*30	ideal three-chain C30 opening
Delta	typed closure loss/correction
3*30 - Delta	effective remaining opening
pi/(30*1260)	carrier x full-closure normalization

Numerical comparison

external check only - not input

alpha_RT	0.0072973525630485
alpha_inv	137.035999201092
CODATA inv	137.035999177(21)
diff inv	2.409e-08

The gate value is computed from the 21|30 ledger. CODATA is displayed only after the computation.

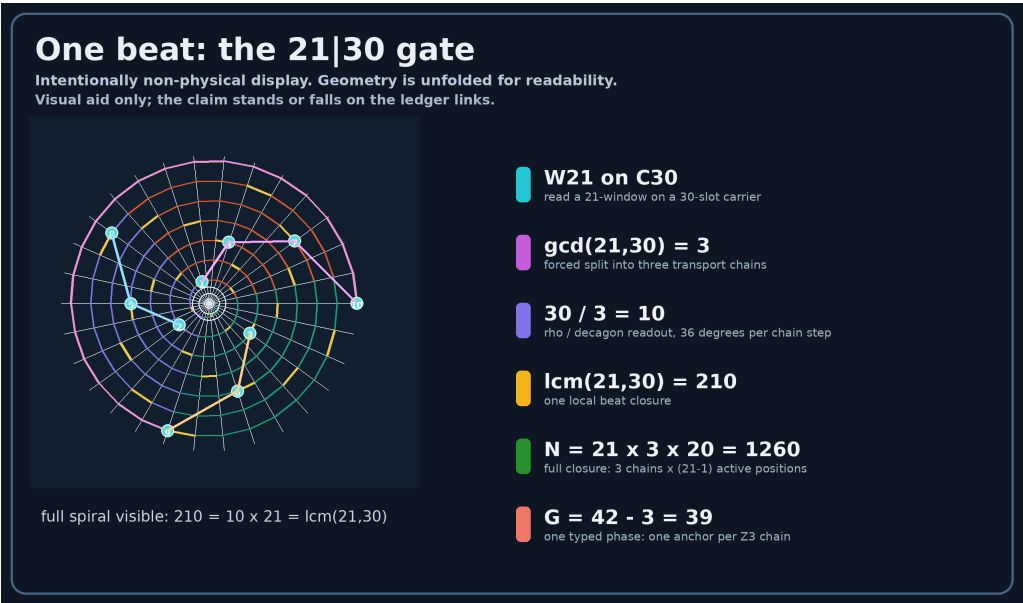
Gate ledger leading to the factors

Link	Consequence
21 30	primitive readout on C30 carrier
3 = gcd(21,30)	three-chain split; source of the 3 in 3*30
10 = 30/3	rho=10 readout scale used in Delta
210 = lcm(21,30)	one beat cell: 7*C30 = 10*21
1260 = 6*210 = 30*42	first full active closure and normalizing scale
G = 42-3 = 3*(14-1) = 39	one A/B-resolved phase reserves one anchor per Z3 chain
D = (21-1)/21 = 20/21	one boundary slot reserved inside carrier-derived 21-domain

Break condition: invalid link, non-unique gate, broken type rule, or equal-or-stricter competing grammar.

Gate detail - carrier-beat and Delta grammar

Six 210-cells split as A/B x Z3; this supports G=39 and the 21-domain for D.



Companion visualizer (not part of the claim): RT_ALPHA_READOUT_GEOMETRY.html. It is an interactive reader aid; the claim stands or falls on ledger links.

Carrier-beat support lemma

Full closure contains six 210-beat cells. With $\theta(n)=360^\circ \times n/1260$, each 210-boundary advances 60 degrees. A three-arm carrier therefore splits the six cells into two complementary triads.

j	$n=j \times 210$	$q=n/30$	phase	class
0	0	0	0 deg	A
1	210	7	60 deg	B
2	420	14	120 deg	A
3	630	21	180 deg	B
4	840	28	240 deg	A
5	1050	35	300 deg	B

$A = \{0,14,28\}$, $B = \{7,21,35\}$ in C30 index
 $1260 = 6 \times 210 = (A/B) \times Z3 \times 210$
 $G = 42 - 3 = 3 \times (14 - 1) = 39$
AB=2 already counts the complementary phases; G counts one typed phase, so 42-3 not 42-6.
Each phase spans 3 beat cells; each cell has $210/30=7$ C30 slots -> $3 \times 7=21$.

Correction grammar and type rule

$\Delta = AB + AB/\rho - 1/(\rho G) - D/(\rho G C S)$

Expansion: $\Delta = 2 + 2/10 - 1/(10 \times 39) - (20/21)/(10 \times 39 \times 30 \times 7)$.
Type rule: each term is measured in its own domain; numerator schema is AB, AB, 1, D.
Counting measure gives numerator 1; readout-conditioned measure gives numerator D.
Active duty: one typed phase exposes $3 \times 7=21$ slots; reserve one -> $D=20/21$.
 $\alpha_{RT}^{-1} = 137.035999201092$; CODATA is external comparison, not input.

Scope and break protocol

Alpha claim only. To refute: break a named gate link, the active-duty derivation, the type rule, or provide an equal-or-stricter competing primitive gate. Free recombinations do not count.

Delta grammar - typed bookkeeping

Two port costs and two returns make the correction readable without adding a separate document.

Delta as typed bookkeeping

Read Delta as bookkeeping, not as four free terms: two port costs are paid first; two returns then give back over-counting at coarser and finer readout scales.

$$\Delta = AB + \frac{AB}{\rho} - \frac{1}{\rho G} - \frac{D}{\rho GCS}$$

The four Delta terms

Term	Value	Bookkeeping role	Simple mechanical picture
AB	2	port cost	opening + closing: two main flanks of the gate
AB/rho	+2/10	internal port cost	the same port read on the internal rho=10 scale
-1/(rho G)	-1/(10*39)	coarse return	one over-counted corner in the gap domain is given back
-D/(rho G C S)	-(20/21)/(10*39*30*7)	fine return	a final returned margin through the full 21 30 readout mask

Sign rule in words: + + pays the two gate costs; - - returns the counted overlap at two resolutions.

The factors inside Delta

Symbol	Value	Structural role	Where it comes from / picture
AB	2	polarity	the port has two flanks: opening + closing
rho	10	internal readout scale	rho=30/3; the C30 carrier reduced by the three-chain split
G	39	gap domain	42-3=3*(14-1): 42 C30 slots minus three carrier anchors
D	20/21	active duty	a 21-domain with one boundary/P-ARM slot reserved
C	30	carrier base	the C30 readout grid
S	7	beat-cell length	210/30=7 C30 positions per 210-cell

Why this page exists

This page is explanatory, not an extra rule. It names the existing Delta grammar as typed bookkeeping so the reader sees the structure before judging the formula as a free numerical fit.